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Research Paper

Sony Interactive Entertainment

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Sony Interactive Entertainment Cybersecurity Management Research

Sony Interactive Entertainment (SIE) will be the subject company for this research in examining their cybersecurity management practices, shortcomings, and present ways to mitigate and resolve these issues. SIE is a multinational corporation with infrastructure all over the world that helps to support the company's video game lineup and overall digital entertainment products. Primarily SIE is responsible for the PlayStation console lineup, a variety of video games through their PlayStation Studios (PSS) development companies, and online services for millions of users. The reason for choosing this company is to highlight its well-known security incidents and examine the cybersecurity management strategies present within a large corporation and present theorized solutions to the issues that they face and analyze the outcomes that came from the resolution of those incidents. For the sake of scope, SIE will be examined as a separate and individual corporation. This is because certain aspects of cybersecurity management and organizational processes in the real world are affected by the needs of its parent company Sony Group Corporation. Therefore, many of their decisions and policies are likely decided with a wider scope and resource allocation in mind even though the challenges and issues faced by SIE are different due to their products and services being unique among other companies under the Sony name.

Cybersecurity Practices Implemented by SIE.

First, by discerning and defining the basics of the assets and online services within the company that are critical to operations we can illustrate an overview of the current security practices that are being implemented. SIE produces all hardware for the PlayStation consoles and manages the software development of its videogame projects through PSS. That is around

100,000 employees not including the supply chains that support operation on those fronts. Additionally, a major player in the realm of SIE is their online services through PlayStation Network (PSN). As of 2023 PSN has over 105 million active unique user accounts. PSN services are multiplatform and support a multitude of features that all have their features and requirements. (SIE,2023) The PlayStation Store is an online marketplace and digital media repository with additional features through PlayStation Plus and PlayStation Extra. Cloud-based on demand media streaming and gaming is available with PlayStation Premium. While PlayStation Music offers audio streaming and management. Many of these features and services for the user offer subscriptions that deal with handling user financial information and personally identifiable information such as name, birthdate, and social security number with the PlayStation credit card services.

The PlayStation online ecosystem is a complex and intricate array of systems and subsystems that support these services offered to the user. As a large-scale multinational corporation SIE employs corresponding cybersecurity management principles in its handling and operation. The backbone of all user-facing features is PSN, which is the massive geographically dispersed network of servers that support day-to-day operations, financial transactions, user logins, and digital services that occur on the network. These servers are located in physically secure data centers. There are 11 large server clusters that service different regions of the world. (Kovacs,2023). The distributed nature of the network is to accommodate outages from natural disasters or denial of service attacks. This is part of a strong disaster recovery and business continuity plan as the network status for PlayStation is of critical importance and making sure the network is resilient to these issues is a primary objective of these policies. This helps maintain the availability of data and resources for users on the network. Recovery time objective

(RTO) is maximum time a corporation can tolerate for recovering to normal operations, while recovery point objective (RPO) is the maximum amount of data an organization can afford to lose in terms of timeframe. As such a large corporation SIE would more than likely be able to shoulder the financial loss unlike some smaller organizations, however due to the size of PlayStation network with on average over 50 million software sales in a single business quarter, keeping these values as low as possible are of critical importance.

As mentioned above PSN supports a large volume of digital sales in a fiscal year and this is all done through the PlayStation store interface. Software membership programs offer other unique aspects through this store and often help users maintain credit card information for subscriptions and allow access for streaming digital products. Other features include maintaining digital libraries for individual users and acting as a repository for software downloads and management. Sony puts a strong emphasis on the protection of user data and other financial information on this front. For the financial information that is saved on the store that data is considered data at rest. This data is stored in tables that are encrypted on PSN (Kovacs,2023). Data in transit which comprises the communication between the console and PlayStation network, and communication between one console to another is also protected with end-to-end encryption standards. It is important to protect the confidentiality and integrity of data as that ensures that only authorized users are allowed to access this data and use and make changes to said information.

In addition to maintaining standards of cybersecurity practices on their IT infrastructure and software applications SIE employs management solutions in the user domain as well. PlayStation Accounts are what users are required to create to interact with all features of the network and are as a result a prime objective for security measures. This act by itself helps

security by requiring new users to verify themselves with a unique email address to help limit dummy accounts and bot accounts. Two factor authentication is also implemented across the network to implement an extra layer of security to keep unauthorized logins from compromising user accounts (PlayStation, 2023). Geolocation is also implemented directly into account creation as users are required to pick a region for their account that is un-editable. This helps prevent external threats from using compromised credentials and logging into a user's account from outside of the country or geographic region. (PlayStation, 2023).

From another angle, Sony also employs Bug Bounty Programs to help the process of hardening the security of PlayStation Network. These are reward-based competitions that the target company engages in, often with the assistance of third-party management companies. Monetary rewards are given to ethical hackers outside of the company if they are able to uncover vulnerabilities in any of PlayStation's assets and present them to the program managers. Often more money is given for the severity of the vulnerability so this offers a great way for companies to test their security configurations in a live environment that could uncover issues that might go unnoticed in a test environment. Additionally, the company is allowed to pick the scope of desired vulnerability to keep activities relevant and current with business direction. For example, in Sony's case their bug bounty program is only interested in security vulnerabilities in the PSN and the supported PS4 and PS5 models, but not any legacy systems. (HackerOne,2020)

Automation is also an important factor in cybersecurity management and SIE utilizes AI-based software to accomplish this. They have detectors that will examine user behavior to flag accounts for suspicious trends in that behavior that deviates from standard activities. This allows for accounts to be locked, purchases denied, or logins rejected in real-time. Communication can then be sent to the user to advise them on account activity or recovery processes. This is a

continuous monitoring solution that allows for this faster response time, which is important in containing compromised accounts and mitigating risks. (PlayStation,2023)

Cybersecurity Incidents and Management issues

The next vector in the analysis of SIE cybersecurity management for their PlayStation product is looking at incidents that have occurred at the company and detailing their incident response and recovery process. These incidents will center around PlayStation Network as the service has been the subject of many security breaches over the years; one of the most infamous incidents being the 2011 hack. Another example that will be contrast is a very recent compromise of employee data that will be used to compare if and how their response process has changed.

Between April 17th, 2011, and April 19th, 2011, there was unauthorized access to the PSN servers that had caused a breach in information of over 77 million account users (Guardian,2011). This was nearly the entirety of every user account active on the network at the time. The data compromised included personally identifiable information (PII) such as names, date of births, passwords, usernames, and other personal information (Guardian,2011). After detecting the intrusion, Sony paused their server activity and left them unavailable to the public for an indeterminate amount of time while they attempted to contain and research the extent of the breach. Servers were only launched again in stages, and this was nearly a month after initial estimates of downtime. In the realm of public relations this became an issue as Sony was reluctant to admit the breach occurred in the first place and the timeline to informing users of compromise to vital parts of their data was not concise and often contradictory (Guardian,2011). During the time of recouping their servers, protecting their data, and reporting the initial breach,

they partnered with outside and internal security teams to diagnose the issue through digital forensics (Guardian,2011). SIE used cyber forensic tactics to identify the originating cause of the breach, which was injection of SQL malware and social engineering. (Shambhavi, 2023)

Locating the abnormal activity on the servers and other server ecosystems was ultimately what led them to the causes of the breach in their system. Once the opening was created, the hackers could take the information they wanted to exploit for other purposes. By diagnosing and locating these exploits Sony and the cyber forensic teams were able to stop the information from leaking any further and to bolster their system defenses to keep another form of this hack from happening.

This analysis was broken down into stages that Sony used to conclude the breach. The main components of this analysis were malware usage, network traffic, and memory forensics (Shambhavi, 2023). By locating the bad malware, they were able to break down the information and discern that the SQL injection used multiple vectors and different communication patterns to infiltrate the server (Shambhavi, 2023). Which led to them being able to analyze the pathways the injection took to determine the timeframe of the data transfers and its connection to unrecognized IP addresses. They utilized the contents of RAM to analyze the memory to then “extract the volatile data from the servers” (Shambhavi, 2023) This was able to shed light on potential evidence and artifacts left behind from using the servers. These artifacts contained “event logs, registry entries, and attempts to cover” the hackers' usage of the servers (Shambhavi, 2023). The attempts were futile after discovering the hackers’ methodology and breach that Sony was able to stop the information from further leaking.

Another example, SIE experienced a recent hack between May 28th, 2023, and May 31st, 2023, that led into June as well (Infotech,2023). This breach was not a compromise of public

information but of Sony's own employees. Sony released a new patch for a program that they have been working on called MOVEit Transfer. And on the same day as this patch a group now known called Cl0p were the perpetrators (Infotech,2023). During this system breach Sony announced to former and current employees of this breach that it estimated that 6,800 employees' personal data had been compromised (Infotech,2023). MOVEit Transfer is a platform that is supposed to provide security for the transfer and exchange of files between servers and even expands to the transfer of data between organizations (Sony Group Portal, 2023). This is a textbook case of a supply-chain attack as the vulnerabilities of this third-party software were used to gain unauthorized access to Sony's systems. This is a new platform that SIE has begun to use to follow a new corporate workflow implementation (Sony Group Portal, 2023). The new workflow includes SIE expanding the use of other platforms to provide ease to consumers.

SIE released information about the incident on June 2nd, 2023, when the breach was identified by a third-party company that they partner with that resides in the United States (Cybernews,2023). The information compromised was that of the SIE employees' Social Security Numbers (SSN) (Cybernews,2023). In response to the breach for the employees SIE has provided support ensuring the monitor credit scores and any chances of identity theft. On the aspect of technology there was evidence of unauthorized downloads on the platform from the points of the vulnerabilities (Infotech,2023). There has not been a release about the steps that were taken in this case of the recent attack, but we can infer that it was like the hack back in 2011. SIE made the initial report and partnered with digital forensic teams to conclude that this was an organized attack by a Russian led cybercrime syndicate (Cybernews,2023). SIE was also able to determine that the attack was only on the MOVEit Transfer platform located in Japan

which during that time was being used for internal testing, which is why only employee data was compromised (Infotech,2023).

These two security incidents are more than a decade apart but share some commonalities that can highlight issues in Sony's cybersecurity policies and incident response process. Poor data management is the first issue; in the 2023 incident the compromised data was extrapolated by the fact that much of it was not encrypted. Additionally, personally identifiable information was stored together with this lack of protection, such as salaries, health information, and PII being in the same directory. Password-protected files were often stored directly with the passwords associated with them and neither were encrypted. The attack also went undetected for several days before Sony became fully aware, similar to the data breach in 2011. Unauthorized access incidents are a major problem in a network that handles as many financial transactions as PlayStation network does. These issues are also perpetuated by a lack of employee training that can stem from a lack of exposure or role rotation to reinforce security requirements and best practices. The other major issue that can be highlighted in Sony's incident response plan is very inconsistent actions from the public relations (PR) team. In the 2011 network data breach the true extent of data compromise was not known and the timeframe for recovery was severely underestimated. This caused a souring in public relations as growing unrest in the user base began to mount because of the lack of clear communication. Though the 2023 breach consisted mainly of employee data and other proprietary information, the delay in detection combined with the delay in notifying said employees lessens public trust in Sony to handle sensitive information in a secure manner.

Management Solutions and Recommendations

The largest vector for the vulnerabilities that SIE has to manage, PlayStation network, has only expanded since the early days and handles a wider array of devices than it ever did in 2011. Security management strategies for large companies need to be robust enough to deal with the scale of incidents that they will experience and handle vulnerabilities in an effective manner.

The first concept that can address some of the security issues in SIE infrastructure is sufficient data management and classification. Classifying data begins with risk assessment and the goal is to determine the nature of data accessible on the network and the possible risk that could arise from that data being compromised. Data classification can be the responsibility of the owner of the data, a data custodian/data steward, or in some cases an automated system can assist in the process. After classification an appropriate management solution can be implemented to secure the data. This is important because different classes of data are subject to different regulatory standards. For instance, financial information is often subject to the Payment Card Industry Data Security Standard (PCI-DSS) which requires that cardholder data must be protected and stored differently based on the required level of protection (Spirion,2023). In SIE case their large quantities of data would be unreasonable to manually classify and couldn't function off purely user-based classification. Therefore, an automated solution with oversight and regular audits to ensure accuracy of the classification could be more beneficial. Overall ensuring that this process is maintained with regular audits and checks will help mitigate the level of damage that can occur from security incidents and breaches. For example, the exposed sensitive information in Sony's 2023 could have been lessened and the impact mitigated with the improved data security that comes from classification and organizational data management.

For a large multinational company, the PR side of the incident response process can be vitally important in managing the expectations of users and stakeholders alike. In the incidents outlined above Sony's PR was lacking. The 2011 data breach of PlayStation network alone highlights this issue in microcosm, as communication was delayed, misleading and often time non-existent on the reality of the event and the path towards resolution. Mitigating weak PR can be addressed by solving issues in communication between PR teams and other parts of the incident response team. This can also come from a lack of leadership and direction. A proper communication framework to efficiently spread information in addition to an active chief information officer (CIO) to drive a single unified message both internally and externally. This will help the PR team give up to date and relevant communication to all parties before, during, and after security incidents.

Improvements to employee training can also support these solutions. Efficient and effective internal communications to help to ensure that proper data management practices are being implemented as policies are taught and enforced. Job and task rotation programs can help employees be more well rounded in all aspects of the process which serves to help strengthen the incident response team.

Conclusion and Observations

Overall, Sony Interactive Entertainment and the products and services under their brand exist as a large multinational corporation with a geographically dispersed IT infrastructure that spans the globe and online services that support millions of concurrent users and activities per day. As a result, their cybersecurity management is a textbook example of how a large company must be managed and benefits and shortcomings that are a part of that. Incident response and

business continuity teams have ample resources and disasters recovery processes are generally handled within a timely manner. However, poor organization can be a downfall as proper data management policies are not followed and complicated supply chains can allow incidents to go unnoticed and security risks to be exacerbated. Additionally poor communication between large often geographically dispersed teams can lead to poor PR that worsens an incident even further by setting up false and unrealistic expectations for consumers and shareholders. Ultimately these issues can be mitigated by utilizing the positives of a large corporation. The overhead and costs of better data management practices and more frequent audits and policy checks outweigh the severity of data risk and compromise during an incident. Additionally, PR communications and actions can be taken to further extremes in a company that can support them. For instance, in the 2023 data breach Sony offered free credit monitoring services and identity restoration services to individuals affected by the compromise of their personal information (Infotech,2023). Moves like this are step in the right direction and the outcome can often far outweigh the cost for a company that is a household name and frequently in headlines.

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